

Problem 5 – Smart Library Management System

Difficulty

★ ★ Beginner

Overview

A small community library wants to organize its book borrowing process. Instead of keeping track of books on paper, the librarian wants a simple application that allows visitors to borrow and return books while keeping the available inventory up to date.

Your task is to build a Streamlit application that helps the librarian manage books throughout the day.

Mission (Streamlit Only)

1. Store the library books.

Store the available books using a dictionary.

Book	Available Copies
Python Basics	5
Data Science	3
Machine Learning	2
Web Development	4

Use Session State so that book quantities remain updated while the application is running.

2. Display the library information.

Your application should display:

- Library title
 - Library description
 - Available books
 - Number of copies for each book
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3. Choose an action.

Allow the librarian to choose one of the following actions using a Selectbox:

- View Books
 - Borrow Book
 - Return Book
-

4. View Books

Display all books and their available copies.

Example

Python Basics : 5 copies

Data Science : 3 copies

Machine Learning : 2 copies

Web Development : 4 copies

5. Borrow a Book

Allow the user to:

- Select a book
- Press a button to borrow it

If the selected book has at least one available copy:

- Decrease the number of copies by 1
- Display a success message

If no copies are available:

- Display an error message

If only one copy remains after borrowing, display a warning message.

6. Return a Book

Allow the user to:

- Select a book
- Press a button to return it

If the book exists:

- Increase the number of copies by 1
 - Display a confirmation message
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7. Keep Daily Statistics

Track throughout the session:

- Total borrowed books
- Total returned books

Display these statistics on the page.

Bonus Mission ★

Track every transaction performed during the session.

Display a history similar to:

Borrowed Python Basics

Returned Data Science

Borrowed Machine Learning

Useful Hints

- Store books using a dictionary.
 - Store the dictionary inside `st.session_state`.
 - Create separate functions for:
 - View Books
 - Borrow Book
 - Return Book
 - Validate that enough copies are available before borrowing.
 - Update the dictionary values after every successful transaction.
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Required Streamlit Components

Text Functions

Use:

- `st.title()`
 - `st.header()`
 - `st.subheader()`
 - `st.write()`
 - `st.divider()`
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Input Widgets

Use:

- `st.selectbox()` → Choose an action
 - `st.selectbox()` → Select a book
 - `st.button()` → Perform the action
-

Status Elements

Use at least four of the following:

- `st.info()`
 - `st.success()`
 - `st.warning()`
 - `st.error()`
-

Layout Requirements

Sidebar

Display:

- Library Rules
 - Working Hours
 - Total Available Books
-

Two Columns

Left Column

- Action selection
- Book selection
- Action button

Right Column

Display:

- Selected book information
 - Remaining copies
 - Action result
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Form

Place all controls inside a form.

The selected action should only be executed after pressing the Submit button.

Expected GUI

 **Smart Library Management System**

Sidebar

Library Rules

- Return books on time
- Handle books carefully

Working Hours

9:00 AM - 6:00 PM

Choose Action

[Borrow Book ▼]

[Select Book ▼]

[Python Basics ▼]

[Submit]

Book Details

Book: Python Basics

Available Copies : 4

 **Borrow completed successfully!**

Statistics

Borrowed Books : 3

Returned Books : 2

Borrowing History

Borrowed Python Basics

Returned Data Science

Borrowed Machine Learning

Python Topics Covered

- **Variables**
 - **Strings**
 - **Integers**
 - **Dictionaries**
 - **Arithmetic Operators**
 - **Assignment Operators**
 - **Comparison Operators**
 - **Conditional Statements (if, elif, else)**
 - **User-defined Functions**
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Streamlit Topics Covered

Displaying Text

- **st.title()**
- **st.header()**
- **st.subheader()**
- **st.write()**
- **st.divider()**

Input Widgets

- **st.selectbox()**
- **st.button()**

Status Elements

- **st.info()**
- **st.success()**
- **st.warning()**
- **st.error()**

Layout

- **st.sidebar**
 - **st.columns()**
 - **st.form()**
 - **st.session_state**
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Learning Objectives

After completing this project, you should be able to:

- **Store and update data using dictionaries.**
- **Persist data using `st.session_state`.**
- **Build an interactive inventory management system using Streamlit.**
- **Organize code into reusable functions.**
- **Design a user-friendly interface using forms, columns, and a sidebar.**
- **Track and display application statistics and transaction history within a Streamlit session.**